

MARCO FERNSTAEDT

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GitHub: <https://github.com/MarcoFernstaedt> | Portfolio: <https://se-portfolio-site.vercel.app> | LinkedIn: <https://www.linkedin.com/in/marco-f-19a372219>

SOFTWARE DEVELOPER

SUMMARY

Software developer focused on React, Next.js, TypeScript, Node.js, Express, MongoDB, API integrations, AI enabled applications, and production web delivery. Built full stack projects with authentication, WebSockets, file uploads, OpenAI Vision and text to speech, Twilio webhooks, Vercel deployment, SEO, analytics, and responsive user interfaces. Customer service background supports clear documentation, user focused debugging, and direct communication with clients and teams.

CORE SKILLS

JavaScript, TypeScript, React, Next.js, Vite, Node.js, Express, FastAPI, MongoDB, Mongoose, REST APIs, WebSockets, Socket.IO, JWT authentication, Clerk, Stream Video, Stream Chat, Monaco Editor, Piston API, OpenAI APIs, Twilio, Resend, Cloudinary, Tailwind CSS, HTML, CSS, Git, GitHub, Vercel, Linux VPS, Nginx, systemd, DNS, SSL, accessibility, debugging, documentation

SOFTWARE DEVELOPMENT EXPERIENCE

Independent Software Developer and AI Systems Builder | Phoenix, AZ | 2025 to Present

- Built and deployed production style software projects across Next.js, React, TypeScript, Node.js, Express, FastAPI, MongoDB, Twilio, OpenAI APIs, Vercel, and Linux VPS infrastructure
- Delivered a production tutoring business website for Saguaro Blossoms Learning with responsive UI, service landing pages, SEO metadata, Schema.org JSON LD, contact intake, Resend email delivery, analytics events, and custom domain deployment
- Built and maintained a public developer portfolio using Next.js, TypeScript, Tailwind CSS, Framer Motion, Vercel Analytics, GitHub metadata APIs, a resume download asset, and an embedded Sentinel project guide
- Developed a FastAPI and Twilio phone agent backend with inbound SMS and voice webhooks, TwiML speech gathering, guarded outbound SMS and call endpoints, environment based configuration, and JSONL event logging
- Maintained Linux VPS based AI agent operations including safe cleanup, script verification, service health checks, DNS and SSL support, process supervision, and operational documentation

PROJECTS

AI Image to Audio Accessibility Tool

GitHub: https://github.com/MarcoFernstaedt/image_accessibility_tool

Live: <https://ita-orpin.vercel.app>

- Built a Next.js and TypeScript image accessibility platform that converts uploaded images into concise spoken descriptions for blind and low vision users
- Implemented an OpenAI Vision to text to speech pipeline using image analysis, generated descriptions, MP3 audio responses, and browser Blob playback and download flows
- Engineered secure image ingestion with multipart FormData parsing, MIME validation, 10 MB file limits, base64 Data URL conversion, and Node.js Buffer based audio streaming
- Added accessibility and production controls including semantic labels, aria live status messaging, keyboard friendly UI, Arcjet shield rules, bot detection, hosting IP checks, and token bucket rate limiting

Saguaro Blossoms Learning Client Website

GitHub: <https://github.com/MarcoFernstaedt/cynthia-tutoring-platform>

Live: <https://saguaroblossomslearningservices.com>

- Built a production Next.js, TypeScript, and Tailwind CSS client website for a tutoring business with responsive pages, mobile navigation, bilingual English and Spanish content, pricing sections, and service conversion flows
- Implemented a server side contact API using Next.js Node runtime and Resend, including required field validation, email validation, honeypot spam prevention, HTML escaping, owner notification emails, and visitor confirmation emails
- Delivered SEO infrastructure with canonical metadata, Open Graph and Twitter previews, sitemap and robots generation, service specific landing pages, and Schema.org data for organization, local business, services, person, and FAQ entities
- Added Vercel Analytics event tracking for language switches, consultation clicks, service clicks, pricing views, navigation, SEO topic clicks, contact form start, and contact form submit events

Real Time Messaging Platform

GitHub: https://github.com/MarcoFernstaedt/socketio_chat_app

- Developed a full stack real time chat application with React, TypeScript, Node.js, Express, MongoDB, Mongoose, and Socket.IO for authenticated one to one messaging
- Implemented secure authentication with bcrypt password hashing, JWT HTTP only cookies, protected REST endpoints, safe user serialization, and authenticated WebSocket handshakes
- Built real time messaging infrastructure with online presence broadcasts, direct message socket delivery, connection handling, disconnection handling, and MongoDB backed conversation history
- Created a responsive chat frontend using Zustand state management, optimistic UI updates, image message support through Cloudinary, notification sounds, contacts filtering, and chats filtering

InterviewLab Code Live Platform

GitHub: https://github.com/MarcoFernstaedt/code_live_platform

- Built a live coding interview platform with React, TypeScript, Express, MongoDB, Clerk authentication, Stream Video, Stream Chat, and Monaco Editor
- Designed protected session APIs for creating, joining, listing, and ending coding sessions with host and participant authorization, active and completed lifecycle state, and MongoDB persistence
- Integrated Stream Video and Stream Chat so each coding session provisions a dedicated video call and chat channel, adds participants on join, and tears down resources when the host ends the session
- Implemented an in browser coding workspace with Monaco Editor, resizable panels, JavaScript, Python, and Java support, and Piston API code execution with compile error, runtime error, and timeout handling

AI Employee Demo Video Suite

- Built a Remotion and React programmatic video generation suite that renders narrated 1280 by 720 MP4 demo videos and preview stills for AI employee sales workflows
- Implemented reusable visual components for workflow demos, including chat simulations, dashboard cards, approval states, process grids, CTA screens, and audio synced scene sequencing
- Added Node based contract tests that verify required compositions, assets, and text while blocking unsafe marketing claims such as guaranteed ROI, staff replacement, and hard coded pricing

EDUCATION AND TRAINING

Self directed software development training through production style projects in React, TypeScript, Next.js, Node.js, MongoDB, authentication, APIs, real time systems, AI integrations, accessibility, and deployment workflows

CompTIA Network Plus in progress